

Appendix 1

Schedule of Activities: Days 1 and 2

	Screening ^{a, b}	Baseline			
Study Day	–2 to 0	1	2		
Time Post Initial Treatment (Assessment Window)		0 Pre-dose (– 4 hrs)	15 min After end of infusion (+ 1 hr)	24 hrs (± 4 hrs)	36 hrs (± 4 hrs)
Informed consent	x				
Inclusion/exclusion criteria	x	x			
Demographic data	x				
Randomization		x			
Medical history		x			
Complete physical examination ^c	x				
Weight		x			
COVID-19 diagnosis ^d	x				
Chest X-ray/CT scan ^e	x				
ECG	x				
Pregnancy test ^f	x				
PaO ₂ /FiO ₂ ^g	x	← Optional →			
SpO ₂ ^h	x	x	x	x	x
Vital signs ^h	x	x	x	x	x
Ordinal scoring ⁱ		x		x	
Adverse events ^j		x		x	
Concomitant medications ^k		x		x	

Appendix 1: Schedule of Activities: Days 1 and 2 (Cont.)

	Screening ^{a, b}	Baseline			
Study Day	–2 to 0	1	2		
Time Post Initial Treatment (Assessment Window)		0 Pre-dose (–4 hrs)	15 min After end of infusion (+1 hr)	24 hrs (±4 hrs)	36 hrs (±4 hrs)
Hematology ^l	x	x		x	
Chemistry ^m	x	x		x	
Study drug administration ⁿ		x			
Central Labs					
Serum PD (CRP, IL-6, sIL-6R)		x ^o	x ^o	x	x
Serum PK ^p		x ^q	x ^q	x	x
Serum sample for exploratory biomarkers		x		x	
SARS-CoV-2 viral load ^r		x		x	
Serum SARS-CoV-2 antibody titer		x			
Cryopreserved PBMCs ^s		x		x	
Whole blood in PAXgene [®] tubes for RNA analyses ^t		x			

CRP = c-reactive protein; CT = computed tomography; ECG = electrocardiogram; eCRF = electronic case report form; NCI CTCAE = National Cancer Institute Common Terminology Criteria for Adverse Events v.5.0; NEWS2 = National Early Warning Score; PaO₂/FiO₂ = arterial oxygen partial pressure/fraction of inspired oxygen; PBMCs = peripheral blood mononuclear cells; PK = pharmacokinetic; PRO-CTCAE = NCI Patient-Reported Outcomes Common Terminology Criteria for Adverse Events; SpO₂ = peripheral capillary oxygen saturation.

Note: On treatment days, all assessments should be performed prior to dosing, unless otherwise specified.

^a Results from standard-of-care tests or examinations performed prior to obtaining informed consent and within 48 hours before randomization may be used; such tests do not need to be repeated for screening.

Appendix 1: Schedule of Activities: Days 1 and 2 (Cont.)

- ^b Informed consent must be documented before any study-specific screening procedure is performed.
- ^c A complete physical examination, performed at screening and per the investigator's discretion during the study, includes an evaluation of the head, eyes, ears, nose, and throat, and the cardiovascular, dermatologic, musculoskeletal, respiratory, gastrointestinal, genitourinary, and neurologic systems. Any abnormality identified at screening should be recorded on the General Medical History and Baseline Conditions eCRF. New or worsened clinically significant abnormalities identified during the study should be reported as adverse events.
- ^d COVID-19 test (SARS-CoV2 PCR) to confirm diagnosis should be performed within 7 days of randomization.
- ^e Screening chest X-ray or CT scans should be performed within 48 hours prior to randomization. If additional chest X-rays/CT scans are taken per local practice during the study, this information should be provided in the eCRF.
- ^f For women of childbearing potential, including those who have had a tubal ligation, positive urine test results will be confirmed with a serum pregnancy test. Study drug infusion must not be administered unless the serum pregnancy test result is negative.
- ^g If arterial blood gases are measured.
- ^h All vital sign measurements (i.e., respiratory rate, pulse rate, systolic and diastolic blood pressures, and body temperature), oxygen saturation and NEWS2-specific assessments (i.e., consciousness and presence or absence of oxygen support) should be recorded together twice daily with approximately 12 hours in between while the patient remains hospitalized. If measured more than once during a 12-hour period, the worst values (highest temperature, respiratory rate, and heart rate; lowest blood pressure, oxygen saturation, and consciousness level) during that period should be recorded on the eCRF.
- ⁱ Assessment of clinical status using the ordinal scale should be recorded at baseline on Day 1/Visit 1 then again daily every morning (between 8 am and 12 pm) for patients who remain hospitalized.
- ^j After informed consent has been obtained but prior to initiation of study drug, only serious adverse events caused by a protocol-mandated intervention should be reported. After initiation of study drug, all adverse events will be reported until 60 days after the final dose of study drug. After this period, the Sponsor should be notified if the investigator becomes aware of any serious adverse event that is believed to be related to prior study drug treatment (see Section 5.6).
- ^k Medication (e.g., prescription drugs, over-the-counter drugs, vaccines, herbal or homeopathic remedies, nutritional supplements) used by a patient in addition to protocol-mandated treatment from 7 days prior to initiation of study drug to the study completion/discontinuation visit.
- ^l Hematology includes WBC count, RBC count, hemoglobin, hematocrit, platelet count, differential count (neutrophils, eosinophils, basophils, monocytes, lymphocytes), and lymphocyte subsets (T cells, B cells, and NK cells; if the test is available at the site).
- ^m Chemistry panel (serum or plasma) includes bicarbonate or total carbon dioxide (if considered standard of care for the region), sodium, potassium, chloride, glucose, BUN or urea, creatinine, total protein, albumin, phosphorus, calcium, total bilirubin, ALP, ALT, and/or AST, uric acid, LDH, ferritin, and D-dimer.

Appendix 1: Schedule of Activities: Days 1 and 2 (Cont.)

- ⁿ Study drug should be administered after collection of all samples for pharmacodynamic and exploratory biomarker analyses. The initial study drug infusion should be given within 4 hours of randomization. If the clinical signs or symptoms worsen or do not improve (reflected by sustained fever or at least a one-category worsening on the 7-category ordinal scale of clinical status), one additional infusion of blinded treatment of TCZ or placebo can be given, 8–24 hours after the initial infusion.
- ^o On Day 1, CRP, IL-6, and sIL-6R samples should be drawn 0–4 hours before the start of infusion and then within 15 minutes (to 1 hour) after the end of the infusion, on the opposite arm as the infusion. Patients receiving a second infusion of study drug should provide extra samples for CRP, IL-6, and sIL-6R prior to and 15 minutes after the end of the infusion, on the opposite arm as the infusion.
- ^p Patients receiving a second infusion of study drug should provide an extra PK sample prior to and 15 minutes (to 1 hour) after the end of the infusion, on the opposite arm as the infusion.
- ^q On Day 1, PK samples should be drawn 0–4 hours before the start of infusion and then within 15 minutes (to 1 hour) after the end of the infusion, on the opposite arm as the infusion.
- ^r Viral load will be assessed by nasopharyngeal swab. Patients who are intubated and undergo bronchoalveolar lavage will have samples taken for virological assessment. It may only be possible to access one nostril if a patient has a nasogastric tube in place, in which case sites may collect a sample from one nostril only and where possible the same nostril should be used.
- ^s For sites capable of sample collection for analysis of T cells by high-dimensional cytometry.
- ^t The first draw of blood should not be for PAXgene[®] tubes to avoid contact with RNA preservation reagent inside the tube.

Appendix 2

Schedule of Activities: Days 3–28

	Days 3–28 ^a																											Study Completion/ Discontinuation
Study Day	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28		
Chest X-ray/CT scan					x							x							x							x	x	
Vital signs ^b	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	
PaO ₂ /FiO ₂ ^c	← Optional →																											Optional
SpO ₂ ^b	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	
Ordinal scoring ^d	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	
Adverse events ^e	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	
Concomitant medications ^f	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	
Hematology ^g	x				x			x				x							x							x	x	
Chemistry ^h	x				x			x				x							x							x	x	
Central Labs																												
Serum PD (CRP, IL-6, sIL-6R)	x				x							x							x							x	x	
Serum PK					x							x							x							x	x	
Serum sample for exploratory biomarkers	x				x							x							x							x	x	
SARS-CoV-2 viral load ⁱ	x	x	x	x	x			x				x							x							x	x	
Serum SARS-CoV-2 antibody titer																										x	x	
Cryopreserved PBMCs ^j	x				x							x							x							x	x	

Appendix 2: Schedule of Activities: Days 3–28 (Cont.)

	Days 3–28 ^a																											Study Completion/ Discontinuation
Study Day	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28		
Whole blood in PAXgene [®] tubes for RNA analyses ^k	x				x																					x	x	

BAL = bronchoalveolar lavage; CRP = c-reactive protein; CT = computed tomography; ECG = electrocardiogram; eCRF = electronic case report form; NCI CTCAE = National Cancer Institute Common Terminology Criteria for Adverse Events v.5.0; PaO₂/FiO₂ = arterial oxygen partial pressure/fraction of inspired oxygen; PBMCs = peripheral blood mononuclear cells; PK = pharmacokinetic; PRO-CTCAE = NCI Patient-Reported Outcomes Common Terminology Criteria for Adverse Events; SpO₂ = peripheral capillary oxygen saturation..

Note: For patients who have been discharged, all assessments should be performed within ± 3 days of the scheduled onsite visit.

- ^a If patients are discharged from hospital prior to Day 28, follow-up visits should occur twice weekly through Day 28. Patients are encouraged to return for onsite visits at least weekly if possible; other follow-up visits may be conducted by telephone or home visits. Patients should return to the site for a Day 28 visit.
- ^b All vital sign measurements (i.e., respiratory rate, pulse rate, systolic and diastolic blood pressures, and body temperature), oxygen saturation and NEWS2-specific assessments (i.e., consciousness and presence or absence of oxygen support) should be recorded together twice daily with approximately 12 hours in between while the patient remains hospitalized. If measured more than once during a 12-hour period, the worst values (highest temperature, respiratory rate, and heart rate; lowest blood pressure, oxygen saturation, and consciousness level) during that period should be recorded on the eCRF. Following hospital discharge these parameters should be recorded once at each return visit to the clinic. Vital signs and oxygen saturation will not be recorded if follow-up visits are conducted by telephone.
- ^c If arterial blood gases are measured.
- ^d Assessment of clinical status using the ordinal scale should be recorded at baseline on Day 1/Visit 1 then again daily every morning (between 8 am and 12 pm) for patients who remain hospitalized.
- ^e After informed consent has been obtained but prior to initiation of study drug, only serious adverse events caused by a protocol-mandated intervention should be reported. After initiation of study drug, all adverse events will be reported until 60 days after the final dose of study drug. After this period, the Sponsor should be notified if the investigator becomes aware of any serious adverse event that is believed to be related to prior study drug treatment (see Section 5.6).
- ^f Medication (e.g., prescription drugs, over-the-counter drugs, vaccines, herbal or homeopathic remedies, nutritional supplements) used by a patient in addition to protocol-mandated treatment from 7 days prior to initiation of study drug to the study completion/discontinuation visit.

Appendix 2: Schedule of Activities: Days 3–28 (Cont.)

- ^g Hematology includes WBC count, RBC count, hemoglobin, hematocrit, platelet count, differential count (neutrophils, eosinophils, basophils, monocytes, lymphocytes), and lymphocyte subsets (T cells, B cells, and NK cells; if the test is available at the site).
- ^h Chemistry panel (serum or plasma) includes bicarbonate or total carbon dioxide (if considered standard of care for the region), sodium, potassium, chloride, glucose, BUN or urea, creatinine, total protein, albumin, phosphorus, calcium, total bilirubin, ALP, ALT, and/or AST, uric acid, LDH, ferritin, and D-dimer.
- ⁱ Viral load will be assessed by nasopharyngeal swab. Patients who are intubated and undergo bronchoalveolar lavage will have samples taken for virological assessment. It may only be possible to access one nostril if a patient has a nasogastric tube in place, in which case sites may collect a sample from one nostril only and where possible the same nostril should be used.
- ^j For sites capable of sample collection for analysis of T cells by high-dimensional cytometry.
- ^k The first draw of blood should not be for PAXgene[®] tubes to avoid contact with RNA preservation reagent inside the tube.

Appendix 3 Schedule of Activities: After Day 28

				Study Completion
Study Day (Assessment Window)	35 ^a (± 3 days)	45 ^a (± 3 days)	60 (± 3 days)	
Chest X-ray/CT scan			x	
SARS-CoV-2 viral load ^b	x	x	x	
Vital signs ^c	x	x	x	
SpO ₂ ^c	x	x	x	
Ordinal scoring ^d	x	x	x	
Adverse events ^e	x	x	x	
Concomitant medications ^f	x	x	x	
Hematology ^g	x	x	x	
Chemistry ^h	x	x	x	
Central Labs				
Serum PD (CRP, IL-6, sIL-6R)	x		x	
Serum PK	x		x	
Serum sample for exploratory biomarkers	x		x	
Serum SARS-Cov-2 antibody titer			x	

CRP = c-reactive protein; CT = computed tomography; PK = pharmacokinetic; SpO₂ = peripheral capillary oxygen saturation.

^a If patients are unable to return for onsite visits at Day 35 and/or Day 45, these may be conducted by telephone or home visits. Patients should return to the site for a Day 60 Study Completion visit.

Appendix 3: Schedule of Activities: After Days 28 (Cont.)

- ^b Patients who remain in hospital will have viral load assessed by nasopharyngeal swabs; these will be done if there is evidence of on-going infection.
- ^c For patients who remain in hospital, vital sign measurements and NEWS2-specific assessments should be conducted twice daily. Following hospital discharge, these parameters should be recorded once at each return visit to the clinic. Vital signs and oxygen saturation will not be recorded if follow-up visits are conducted by telephone.
- ^d Assessment of clinical status using the ordinal scale should be recorded at baseline on Day 1/Visit 1 then again daily every morning (between 8 am and 12 pm) for patients who remain hospitalized.
- ^e After informed consent has been obtained but prior to initiation of study drug, only serious adverse events caused by a protocol-mandated intervention should be reported. After initiation of study drug, all adverse events will be reported until 60 days after the final dose of study drug. After this period, the Sponsor should be notified if the investigator becomes aware of any serious adverse event that is believed to be related to prior study drug treatment (see Section 5.6).
- ^f Medication (e.g., prescription drugs, over-the-counter drugs, vaccines, herbal or homeopathic remedies, nutritional supplements) used by a patient in addition to protocol-mandated treatment from 7 days prior to initiation of study drug to the study completion/discontinuation visit.
- ^g Hematology includes WBC count, RBC count, hemoglobin, hematocrit, platelet count, differential count (neutrophils, eosinophils, basophils, monocytes, lymphocytes), and lymphocyte subsets (T cells, B cells, and NK cells). Hematology labs will not be performed if follow-up visits are conducted by telephone.
- ^h Chemistry panel (serum or plasma) includes bicarbonate or total carbon dioxide (if considered standard of care for the region), sodium, potassium, chloride, glucose, BUN or urea, creatinine, total protein, albumin, phosphorus, calcium, total bilirubin, ALP, ALT, and/or AST, uric acid, LDH, ferritin, and D-dimer. Chemistry labs will not be performed if follow-up visits are conducted by telephone.